Tomato on the Vine (large)

Overall information

Type of crop	Vine crop
Reference	De Ruiter Seeds
Variety	Merlice
Weight	150-160 gr.(5 per tros)

Growing strategy

Growing cycle	Seasonal/year round
Production cycle	11 months
First production after	8 (summer)-10 (winter) weeks
Origin of youngplants	Seeds
Start cycle	Youngplants propagator/seeds nursery

Points of attention

BER & Bi-coloring

Very strong grower

Ideal climate conditions for top production

		Day	Night
Temperatures	Min	22°C	14°C
	Max	27°C	18°C
Relative humidity	Min	65%	80%
	Max	80%	90%
Daylength	Min	8	*hours*
	Max	18	*hours*
Radiation levels	Min	1000	*j/cm ^{2} /p.day*
	Max	2500	*j/cm ^{2} /p.day*

Points of attention

24H avg. Min. 16°C to 18,5°C - onbelicht/belicht, 24H avg. Max. 20,5°C to 23°C

full grown crop

Technical configuration

Cultivation system	Gutter system
Irrigation	Dripper
Lighting	Yes/No
Heating	Heatblock/Tube rail/grow pipe

Production

	With lighting			Without lighting		
	Plastic	Venlo	UC	Plastic	Venlo	UC
Min. kg/m2		70	70	50	60	60
Max. kg/m2		100	100	65	80	80

UC, waarde uitleggen +1

Operational costs

Labour needed	1 ha	5 ha	10 ha
Low level workers	20000	110000h	200000h
Skilled/semi-skilled workers	10000h	55000h	100000h

need 56 people for 10 ha in the Nederlands

Harvesting and Packing

Harvesting method	manual picking/truss harvest
Internal transport	harvest trollies
Cleaning	not applicable
Packaging	semi-automatic
Type of Package	carton box (6kg)/flow pack/plastic box
Storage	14-15°C

Packaging type

Boxes 5-10 kg, scales 500-750g

No big temp. changes allowed min. 12

Points of attention

Sensitive for viruses	No
Sensitive for Fungi	No

Link to hygiene protocol file

zwavel

Cherry Tomato

Overall information

Type of crop	Cherry truss
Reference	De Ruiter Seeds
Variety	Juanita
Weight	13 gr * 14 fruits per cluster

Growing strategy

Points of attention

Growing cycle	Seasonal/year round
Production cycle	11 months
First production after	8 (summer)-10 (winter) weeks
Origin of youngplants	Seeds
Start cycle	Youngplants propagator/seeds nursery

Ideal climate conditions for top production

Points of attention

		Day	Night
Temperatures	Min	22°C	14°C
	Max	28°C	17°C
Relative humidity	Min	65%	80%
	Max	80%	90%
Daylength	Min	8	*hours*
	Max	18	*hours*
Radiation levels	Min	1000	*j/cm ^{2} /p.day*
	Max	2500	*j/cm ^{2} /p.day*

Technical configuration

Cultivation system	Gutter system
Irrigation	Dripper
Lighting	Yes/No
Heating	Heatblock/Tube rail/grow pipe

Production

	With lighting			Without lighting		
	Plastic	Venlo	UC	Plastic	Venlo	UC
Min. kg/r	0	25	25	18	23	23
Max. kg/ı	0	30	30	23	27	27

Operational costs

Labour needed	1 ha	5 ha	10 ha
Low level workers	mer +30%		
Skilled/semi-skilled workers			

Harvesting and Packing

Packaging type

Harvesting method	manual picking/truss harvest
Internal transport	harvest trollies
Cleaning	not applicable
Packaging	semi-automatic
Type of Package	carton box (3kg)/flow pack/plastic box
Storage	14-15°C

Points of attention

Sensitive for viruses	No
Sensitive for Fungi	Yes

Beef Tomato

Overall information

Type of crop	Beef
Reference	De Ruiter Seeds
Variety	Torero
Weight	250gr (3/4 vruchten per tros)

Growing strategy

Points of attention

Growing cycle	Seasonal/year round
Production cycle	11 months
First production after	8 (summer)-10 (winter) weeks
Origin of youngplants	Seeds
Start cycle	Youngplants propagator/seeds nursery

Ideal climate conditions for top production

Points of attention

		Day	Night
Temperatures	Min	22°C	15°C
	Max	27°C	18°C
Relative humidity	Min	65%	80%
	Max	80%	90%
Daylength	Min	8	*hours*
	Max	18	*hours*
Radiation levels	Min	1000	*j/cm ^{2} /p.day*
	Max	2500	*j/cm ^{2} /p.day*

Technical configuration

Cultivation system	Gutter system
Irrigation	Dripper
Lighting	Yes/No
Heating	Heatblock/Tube rail/grow pipe

Production

	With lighting			Without lighting		
	Plastic	Venlo	UC	Plastic	Venlo	UC
Min. kg/r	0	65	65	45	55	55
Max. kg/ı	0	85	85	60	70	70

Operational costs

Labour needed	1 ha	5 ha	10 ha
Low level workers	mer +10%		
Skilled/semi-skilled workers			

Harvesting and Packing

Packaging type

Harvesting method	manual picking/loose harvest
Internal transport	harvest trollies
Cleaning	not applicable
Packaging	semi-automatic
Type of Package	carton box (6kg)/plastic box
Storage	14-15°C

Points of attention

Sensitive for viruses	No
Sensitive for Fungi	Yes

Cucumber Long English

Overall information

Type of crop	High Wire
Reference	De Ruiter Seeds
Variety	Garpo
Weight	370-380 gr.

Growing strategy

Points of attention

Growing cycle	6 months
Production cycle	5 months
First production after	1 months
Origin of youngplants	Seeds
Start cycle	Youngplants propagator/seeds nursery

Ideal climate conditions for top production

Points of attention

		Day	Night	
Temperatures	Min	21	17.5	
	Max	27	19	
Relative humidity	Min	70%	80%	
	Max	85%	90%	
Daylength	Min	8	*hours*	
	Max	18	*hours*	
Radiation levels	Min	2000	*'cm ^{2} /p.day*	able to handle higher intesity than toma
	Max	3000	*'cm ^{2} /p.day*	

Technical configuration

Cultivation system	Gutter system
Irrigation	Dripper
Lighting	Yes/No
Heating	Heatblock/Tube rail/grow pipe

Production

	With lighting			Without lighting		
	Plastic	Venlo	UC	Plastic	Venlo	UC
st Min. kg/r	220	250	280	180	220	220
st Max. kg/ı	250	320	320	230	270	270

Operational costs

Labour needed	1 ha	5 ha	10 ha
Low level workers			
Skilled/semi-skilled workers			

Harvesting and Packing

Packaging type

Harvesting method	manual picking/loose harvest
Internal transport	harvest trollies
Cleaning	not applicable
Packaging	semi-automatic
Type of Package	carton box (5kg)/flow pack/plastic box/plastic seal
Storage	14-15°C

Points of attention

Sensitive for viruses	Yes	high need for resistance package (CGMMV)
Sensitive for Fungi	Yes	high need for resistance package (PX/PYThiuM

Cucumber Snack/mini

Overall information

Type of crop	High Wire
Reference	De Ruiter Seeds
Variety	Mini star
Weight	40 - 50 gr.

Growing strategy

Points of attention

Growing cycle	6 months
Production cycle	5 months
First production after	1 months
Origin of youngplants	Seeds
Start cycle	Youngplants propagator/seeds nursery

Ideal climate conditions for top production

Points of attention

		Day	Night
Temperatures	Min	21	17.5
	Max	28	20
Relative humidity	Min	70%	80%
	Max	85%	90%
Daylength	Min	8	*hours*
	Max	18	*hours*
Radiation levels	Min	2000	*'cm ^{2} /p.day*
	Max	3000	*'cm ^{2} /p.day*

more generative climate strategy

Technical configuration

Cultivation system	Gutter system
Irrigation	Dripper
Lighting	Yes/No
Heating	Heatblock/Tube rail/grow pipe

Production

	With lighting				Without lighting		
	Plastic	Venlo	UC		Plastic	Venlo	UC
st Min. kg/r	77	88	98	0	63	77	77
st Max. kg/ı	88	112	112	0	81	95	95

Operational costs

Labour needed	1 ha	5 ha	10 ha
Low level workers			
Skilled/semi-skilled workers			

Harvesting and Packing

Packaging type

Harvesting method	manual picking/loose harvest
Internal transport	harvest trollies
Cleaning	not applicable
Packaging	semi-automatic
Type of Package	carton box (5kg)/flow pack/plastic box
Storage	14-15°C

Points of attention

Sensitive for viruses	Yes	high need for resistance package (CGMMV)
Sensitive for Fungi	Yes	high need for resistance package (PX/PYThiuM

Cucumber Long English

Overall information

Type of crop	Traditional (Umbrella)
Reference	De Ruiter Seeds
Variety	Gideon
Weight	400 gr.

Growing strategy

Points of attention

Growing cycle	4 months
Production cycle	3 months
First production after	1 months
Origin of youngplants	Seeds
Start cycle	Youngplants propagator/seeds nursery

Ideal climate conditions for top production

Points of attention

		Day	Night
Temperatures	Min	21	17.5
	Max	27	19
Relative humidity	Min	70%	80%
	Max	85%	90%
Daylength	Min	8	*hours*
	Max	18	*hours*
Radiation levels	Min	2000	*'cm ^{2} /p.day*
	Max	3000	*'cm ^{2} /p.day*

Technical configuration

Cultivation system	Gutter system
Irrigation	Dripper
Lighting	Yes/No
Heating	Heatblock/Tube rail/grow pipe

Production

	With lighting			Without lighting		
	Plastic	Venlo	UC	Plastic	Venlo	UC
st Min. kg/m2				140	170	170
st Max. kg/m2				180	230	230

Operational costs

Labour needed	1 ha	5 ha	10 ha
Low level workers	16600		
Skilled/semi-skilled workers	8300		

Harvesting and Packing

Packaging type

Harvesting method	manual picking/loose harvest
Internal transport	harvest trollies
Cleaning	not applicable
Packaging	semi-automatic
Type of Package	carton box (5kg)/flow pack/plastic box/plastic seal
Storage	14-15°C

Points of attention

Sensitive for viruses	Yes	high need for resistance package (CGMMV)
Sensitive for Fungi	Yes	high need for resistance package (PX/PYThiuM

Cucumber Snack/mini

Overall information

Type of crop	Traditional (umbrella)
Reference	De Ruiter Seeds
Variety	Mini star
Weight	40 - 50 gr.

Growing strategy

Points of attention

Growing cycle	4 months 3 cycles py
Production cycle	3 months
First production after	1 months
Origin of youngplants	Seeds
Start cycle	Youngplants propagator/seeds nursery

Ideal climate conditions for top production

Points of attention

		Day	Night
Temperatures	Min	21	17.5
	Max	28	20
Relative humidity	Min	70%	80%
	Max	85%	90%
Daylength	Min	8	*hours*
	Max	18	*hours*
Radiation levels	Min	2000	*'cm ^{2} /p.day*
	Max	3000	*'cm ^{2} /p.day*

more generative climate strategy

Technical configuration

Cultivation system	Gutter system
Irrigation	Dripper
Lighting	Yes/No
Heating	Heatblock/Tube rail/grow pipe

Production

	With lighting			Without lighting		
	Plastic	Venlo	UC	Plastic	Venlo	UC
st Min. kg/m2						
st Max. kg/m2				20% lager dan hw		

Operational costs

Labour needed	1 ha	5 ha	10 ha
Low level workers	30000		
Skilled/semi-skilled workers	15000		

Harvesting and Packing

Packaging type

Harvesting method	manual picking/loose harvest
Internal transport	harvest trollies
Cleaning	not applicable
Packaging	semi-automatic
Type of Package	carton box (5kg)/flow pack/plastic box
Storage	14-15°C

Points of attention

Sensitive for viruses	Yes	high need for resistance package (CGMMV)
Sensitive for Fungi	Yes	high need for resistance package (PX/PYThiuM

Lettuce Head

Overall information

Type of crop	Head lettuce, butter head
Reference	Rijk Zwaan
Variety	Cecilia
Weight	200 g

Growing strategy

Points of attention

Growing cycle	year-round
Production cycle	35-45 days
First production after	35-45 days
Origin of youngplants	seeds
Start cycle	seeds, transplanting 1-2 times

Ideal climate conditions for top production

Points of attention

see next page

		Day	Night
Temperatures	Min		
	Max		
Relative humidity	Min		
	Max		
Daylength	Min		*hours*
	Max		*hours*
Radiation levels	Min		*j/cm ^{2} /p.day*
	Max		*j/cm ^{2} /p.day*

Technical configuration

Cultivation system	closed MGS
Irrigation	MGS
Lighting	HPS/ LED
Heating	below, upper, gable heatings

Production

	With lighting			Without lighting		
	Plastic	Venlo	UC	Plastic	Venlo	UC
Min. kg/m2		70	80		60	70
Max. kg/m2		90	100		80	90

Operational costs

Labour needed	1 ha	5 ha	10 ha
Low level workers			
Skilled/semi-skilled workers			

no relevant data

Harvesting and Packing

Packaging type

Harvesting method	automatic and manaully
Internal transport	MGS
Cleaning	automatic gutter cleaning and disinfection
Packaging	
Type of Package	
Storage	cold storage 8- 10 degrees

Points of attention

Sensitive for viruses

Sensitive for Fungi

Growth climates			Head lettuce
Temperature (℃)	Day	Optimal	22
		Minimum	15
		Maximum	30
	Night	Optimal	18
		Minimum	10
		Maximum	22
RH (%)	Day	Optimal	70
		Minimum	50
		Maximum	80
	Night	Optimal	85
		Minimum	60
		Maximum	90
Solar radiation (W/m^2)		Optimal	500
		Minimum	400
		Maximum	700
DLI ($mol/m^2 \cdot day$)		Optimal	20
		Minimum	14
		Maximum	30
Day length (h)		Optimal	16
		Minimum	12
		Maximum	18
CO_2 (ppm)		Optimal	800
		Minimum	500
		Maximum	1000

Lettuce Baby leaf & Teen Leaf

Overall information

Type of crop	Lettuce Baby leaf & Teen leaf
Reference	BASF
Variety	Finstar
Weight	14-28 g

Growing strategy

Growing cycle	year-round
Production cycle	21-28 days
First production after	21-28 days
Origin of youngplants	seeds
Start cycle	seeds

Points of attention

Ideal climate conditions for top production

		Day	Night
Temperatures	Min		
	Max		
Relative humidity	Min		
	Max		
Daylength	Min		*hours*
	Max		*hours*
Radiation levels	Min		*j/cm ^{2} /p.day*
	Max		*j/cm ^{2} /p.day*

Points of attention

see next page

Technical configuration

Cultivation system	open moving gutter system
Irrigation	MGS
Lighting	HPS / LED
Heating	under, above, gable heatings

Production

	With lighting		Without lighting	
	Plastic Venlo	UC	Plastic Venlo	UC
Min. kg/m2	65	80	55	70
Max. kg/m2	85	100	75	90

Operational costs

Labour needed	1 ha	5 ha	10 ha
Low level workers			
Skilled/semi-skilled workers			

no relevant data

Harvesting and Packing

Harvesting method	automatic
Internal transport	MGS
Cleaning	automatic gutter cleaning & disinfection
Packaging	automatic
Type of Package	mixed salad bag
Storage	cold storage 8-10 degrees

Packaging type

Points of attention

Sensitive for viruses

Sensitive for Fungi

Growth climates			Head lettuce	Teen leaf
Temperature (℃)	Day	Optimal	22	24
		Minimum	15	15
		Maximum	30	30
	Night	Optimal	18	19
		Minimum	10	10
		Maximum	22	22
RH (%)	Day	Optimal	70	70
		Minimum	50	60
		Maximum	80	85
	Night	Optimal	85	85
		Minimum	60	65
		Maximum	90	90
Solar radiation (W/m^2)		Optimal	500	500
		Minimum	400	400
		Maximum	700	700
DLI ($mol/m^2 \cdot day$)		Optimal	20	22
		Minimum	14	14
		Maximum	30	30
Day length (h)		Optimal	16	18
		Minimum	12	12
		Maximum	18	20
CO_2 (ppm)		Optimal	800	800
		Minimum	500	500
		Maximum	1000	1000

Pepper Block

Overall information

Type of crop	High wire crop
Reference	Rijk Zwaan
Variety	Redline
Weight	190-210 g

Growing strategy

Points of attention

Growing cycle	year-round
Production cycle	11 months
First production after	2-3 months
Origin of youngplants	seeds
Start cycle	young plants from propagation

Ideal climate conditions for top production

Points of attention

		Day	Night
Temperatures	Min	20	17
	Max	25	19
Relative humidity	Min	65%	80%
	Max	80%	90%
Daylength	Min	8	*hours*
	Max	16	*hours*
Radiation levels	Min	1000	*'cm ^{2} /p.day*
	Max	2500	*'cm ^{2} /p.day*

Technical configuration

Cultivation system	V system, double gutters
Irrigation	dripper irrigation
Lighting	HPS
Heating	AHU, pipes

Production

	With lighting			Without lighting		
	Plastic	Venlo	UC	Plastic	Venlo	UC
Min. kg/m2		30	30		20	25
Max. kg/m2		35	40		30	32

The production numbers depends a lot on the climates (temp, humidity, light), which varies from locations to locations

Operational costs

Labour needed	1 ha	5 ha	10 ha
Low level workers			
Skilled/semi-skilled workers			

hours per year around 625/1000 m (including clean-up)

Harvesting and Packing

Packaging type

Harvesting method	manually
Internal transport	Cart
Cleaning	
Packaging	semi-automatic
Type of Package	depends
Storage	

Points of attention

Sensitive for viruses

Sensitive for Fungi

Pepper Sweetpoint

Overall information

Type of crop	High wire crop
Reference	Rijk Zwaan
Variety	Kappy RZ F1
Weight	100 to 200 g

Growing strategy

Points of attention

Growing cycle	year-round
Production cycle	11 months
First production after	2-3 months
Origin of youngplants	seeds
Start cycle	young plants from propagation

Ideal climate conditions for top production

Points of attention

		Day	Night
Temperatures	Min	20	17
	Max	25	19
Relative humidity	Min	65%	80%
	Max	80%	90%
Daylength	Min	8	*hours*
	Max	16	*hours*
Radiation levels	Min	1000	*'cm ^{2} /p.day*
	Max	2500	*'cm ^{2} /p.day*

Technical configuration

Cultivation system	V system, double gutters
Irrigation	dripper irrigation
Lighting	HPS
Heating	AHU, pipes

Production

	With lighting			Without lighting		
	Plastic	Venlo	UC	Plastic	Venlo	UC
Min. kg/m2		15	15		10	13
Max. kg/m2		18	20		15	16

The production numbers depends a lot on the climates (temp, humidity, light), which varies from locations to locations

Operational costs

Labour needed	1 ha	5 ha	10 ha
Low level workers			
Skilled/semi-skilled workers			

hours per year around 625/1000 m (including clean-up)

Harvesting and Packing

Packaging type

Harvesting method	manually
Internal transport	Cart
Cleaning	
Packaging	semi-automatic
Type of Package	depends
Storage	

Points of attention

Sensitive for viruses

Sensitive for Fungi

Strawberry Everbearing

Overall information

Type of crop	Strawberry everbearing
Reference	Genson
Variety	Karima
Weight	25-30 g

Growing strategy

Points of attention

Growing cycle	year-round	
Production cycle	1 year	
First production after	2-4 weeks	
Origin of youngplants	Tray plants, mini tray plants, misted tips	fresh / frozen
Start cycle	youngplants	

Ideal climate conditions for top production

Points of attention

		Day	Night	
Temperatures	Min	20	8	
	Max	26	12	
Relative humidity	Min	60%	85%	
	Max	75%	95%	
Daylength	Min	14	*hours*	
	Max	20	*hours*	
Radiation levels	Min		*j/cm ^{2} /p.day*	DLI of 20 to 30 mol/m2/d
	Max		*j/cm ^{2} /p.day*	

Technical configuration

Cultivation system	gutter system	swing / pyramid systems
Irrigation	dripper line	
Lighting	LED	with far red
Heating	AHU / pipe system	

Production

	With lighting			Without lighting			
	Plastic	Venlo	UC	Plastic	Venlo	UC	without lighting production depends a lo
Min. kg/m2		12	18		8	15	depends on the cultivation system and t
Max. kg/m2		18	25		14	22	

Operational costs

Labour needed	1 ha	5 ha	10 ha
Low level workers			
Skilled/semi-skilled workers			

Harvesting and Packing

Packaging type

Harvesting method	manually harvesting	
Internal transport	rail pipe system and cart	There is a meeting about NADEC strawberry p
Cleaning		
Packaging	depends	
Type of Package	depends	
Storage	cold storage room, and pre cooling	

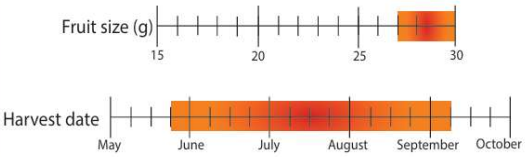
Points of attention

Sensitive for viruses		
Sensitive for Fungi	yes	Sensitive to botrytis

Karima is an everbearing variety from Mattivi's breeding program. The variety is productive and the fruits are very firm, juicy and have a nice taste.

Karima has a perfectly conical shape that has a large size maintained throughout the season. It produces a very juicy fruits with a medium-high Brix degree. The percentage of small fruits is very low. Karima's production is high on the first flush, especially if two or three crown tray plants are used.

The plants have very robust flower stems of medium length. It is easy to pick and develops a good amount of leaves. Karima is tolerant to powdery mildew and botrytis. The plants are not susceptible to root system diseases, but can suffer when plants are placed too deep in the substrate.



ɔt on the outside light level

he plant density

ıackaging line. More info could be add later.

Strawberry June Bearing

Overall information

Type of crop	strawberry June bearing
Reference	Genson
Variety	Sonsation
Weight	20-25

Growing strategy

Points of attention

Growing cycle	3 months
Production cycle	2 months
First production after	2 weeks
Origin of youngplants	Tray plants, mini tray plants, misted tips
Start cycle	youngplants

Ideal climate conditions for top production

Points of attention

		Day	Night
Temperatures	Min	20	8
	Max	22	12
Relative humidity	Min	60%	85%
	Max	75%	95%
Daylength	Min	12	*hours*
	Max	20	*hours*
Radiation levels	Min		*j/cm ^{2} /p.day*
	Max		*j/cm ^{2} /p.day*

DLI of 20 to 30 mol/m2/d

Technical configuration

Cultivation system	gutter
Irrigation	dripper line
Lighting	LED
Heating	AHU / pipe system

Production

	With lighting			Without lighting		
	Plastic	Venlo	UC	Plastic	Venlo	UC
Min. kg/m2		10	10		6	6
Max. kg/m2		16	16		12	12

without lighting production depends a lc
depends on the crop cycles per year

Operational costs

Labour needed	1 ha	5 ha	10 ha
Low level workers			
Skilled/semi-skilled workers			

Harvesting and Packing

Packaging type

Harvesting method	manually harvesting
Internal transport	rail pipe system and cart
Cleaning	
Packaging	depends
Type of Package	depends
Storage	cold storage room, and pre cooling

There is a meeting about NADEC strawberry p

Points of attention

Sensitive for viruses	
Sensitive for Fungi	yes

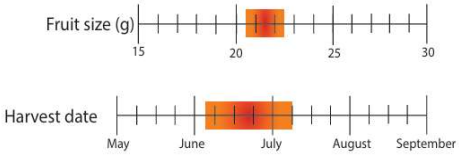
Sensitive to botrytis

Sonsation is a mid-season variety from Flevo Berry's breeding program. The berries are juicy and have an excellent taste, you will eat them in one sitting.

The plants are easy to grow, they are compact and the leaves are upright. The flower clusters are between the leaves and are thus protected against night frost. The flowers produce good-quality pollen which guarantees a good harvest.

The fruits are easy to pick due to its uniform fruits. These have a perfect sweet acid balance and have a beautiful shiny colour which they preserve after picking. The yield is high and the plant produces a high percentage of class I fruits.

Sonsation is tolerant for crown rot *(Phytophthora cactorum)* and has little problems with mildew if grown in the right conditions.



ɔt on the outside light level

ıackaging line. More info could be add later.

Eggplants

Overall information

Type of crop	Eggplant
Reference	Rijk Zwaan
Variety	Beyonce RZ F1
Weight	150 g

Growing strategy

Points of attention

Growing cycle	year round
Production cycle	10 months
First production after	2 months
Origin of youngplants	seeds
Start cycle	youngplant from propogator

Ideal climate conditions for top production

Points of attention

see next page

		Day	Night
Temperatures	Min		
	Max		
Relative humidity	Min		
	Max		
Daylength	Min		*hours*
	Max		*hours*
Radiation levels	Min		*j/cm ^{2} /p.day*
	Max		*j/cm ^{2} /p.day*

Technical configuration

Cultivation system	gutter system
Irrigation	drip system
Lighting	yes /no
Heating	AHU / pipe systems

Production

	With lighting			Without lighting		
	Plastic	Venlo	UC	Plastic	Venlo	UC
Min. kg/m2		50	55		40	45
Max. kg/m2		60	65		50	55

Operational costs

Labour needed	1 ha	5 ha	10 ha
Low level workers			
Skilled/semi-skilled workers			

Harvesting and Packing

Packaging type

Harvesting method	manually
Internal transport	Cart
Cleaning	
Packaging	semi-automatic
Type of Package	depends
Storage	

Points of attention

Sensitive for viruses

Sensitive for Fungi

Roses

Overall information

Type of crop	Rose
Reference	Pligt pot roses
Variety	many different varieties
Weight	different pot size, 10.5 cm, 12 cm, 14 cm

Growing strategy

Points of attention

Growing cycle	year-round
Production cycle	10 weeks
First production after	10 weeks
Origin of youngplants	mother plants
Start cycle	cuttings

Ideal climate conditions for top production

Points of attention

climate condition

see next page

		Day	Night
Temperatures	Min		
	Max		
Relative humidity	Min		
	Max		
Daylength	Min		*hours*
	Max		*hours*
Radiation levels	Min		*j/cm ^{2} /p.day*
	Max		*j/cm ^{2} /p.day*

Technical configuration

Cultivation system	open table system
Irrigation	ebb and flow
Lighting	HPS/LED
Heating	under pipe, above pipe, gable pipe

Production

	With lighting			Without lighting		
	Plastic	Venlo	UC	Plastic	Venlo	UC
Min. kg/m2						
Max. kg/m2						

no relevant info

for cur flowr production, normally 300 r

Operational costs

Labour needed	1 ha	5 ha	10 ha
Low level workers			
Skilled/semi-skilled workers			

for 4 ha GH

50-150 people, depending on season

Harvesting and Packing

Packaging type

Harvesting method	
Internal transport	
Cleaning	
Packaging	
Type of Package	
Storage	

Points of attention

Sensitive for viruses		
Sensitive for Fungi	yes	botrytis and powdery mildew

Growth climates			Rose
Temperature (°C)	Day	Optimal	24-25
		Minimum	19
		Maximum	28
	Night	Optimal	16
		Minimum	15
		Maximum	17
RH (%)	Day	Optimal	75
		Minimum	55
		Maximum	85
	Night	Optimal	85
		Minimum	75
		Maximum	90
Solar radiation (W/m^2)		Optimal	700-1000
		Minimum	-
		Maximum	1000
DLI ($mol/m^2 \cdot day$)		Optimal	24-25
		Minimum	19
		Maximum	30
Day length (h)		Optimal	20
		Minimum	16
		Maximum	24
CO_2 (ppm)		Optimal	800
		Minimum	400
		Maximum	1000

oses per m2 per year

Orchids

Overall information

Type of crop	Phalaenopsis
Reference	Anthura
Variety	Many different varieties (25,000 + species)
Weight	different pot size, e.g 12 cm pot

Growing strategy

Points of attention

Growing cycle	year-round
Production cycle	18 months
First production after	18 months
Origin of youngplants	bottle plants
Start cycle	transplant bottle plants into plugplants in trays

Ideal climate conditions for top production

Points of attention

		Day	Night
Temperatures	Min		
	Max		
Relative humidity	Min		
	Max		
Daylength	Min		*hours*
	Max		*hours*
Radiation levels	Min		*j/cm ^{2} /p.day*
	Max		*j/cm ^{2} /p.day*

For phalaenopsis, there are 3 growing pl
the climate conditions, see next page

Technical configuration

Cultivation system	open table system
Irrigation	ebb and flow
Lighting	HPS/LED
Heating	ondernet/bovennet/gevelverwarming

Production

	With lighting			Without lighting		
	Plastic	Venlo	UC	Plastic	Venlo	UC
Min. kg/m2						
Max. kg/m2						

production number see next page exam

Operational costs

no relevant information

Labour needed	1 ha	5 ha	10 ha
Low level workers			
Skilled/semi-skilled workers			

Harvesting and Packing

Packaging type

Harvesting method	tafel / band
Internal transport	tafels / systeem
Cleaning	nee
Packaging	inhoesen / trays
Type of Package	plastic / karton (/ pot)
Storage	niet (in kas) / evetnueel buffer in verwerkingsruimte

Points of attention

Sensitive for viruses	nee	
Sensitive for Fungi	ja	fusarium, rhizoctonia

Growth climates			Heating	Cooling	Flowering
Temperature (°C)	Day	Optimal	28	19	21
		Minimum	27	18	20
		Maximum	30	22	22
	Night	Optimal	28	18	20
		Minimum	27	17	19
		Maximum	28	22	21
RH (%)	Day	Optimal	75	75	75
		Minimum	60	60	60
		Maximum	85	85	85
	Night	Optimal	80	80	80
		Minimum	60	60	60
		Maximum	85	85	85
LI (µmol/m2/s)		Optimal	171	200	238
		Minimum	152	190	209
		Maximum	190	209	266
SR (W/m2)		Optimal	107	125	148
		Minimum	95	119	131
		Maximum	119	131	166
DLI (mol/m^2·day)		Optimal	7	7	7
		Minimum	5	5	5
		Maximum	8	8	8
Day length (h)		Optimal	14	14	14
		Minimum	12	12	12
		Maximum	16	16	16
CO_2 (ppm)		Optimal	800	800	800
Night optimal		Minimum	400	400	400
Day minimum		Maximum	800	800	800

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Anthura

CULTURE PLANNING	***Phalaenopsis Ø12 cm***			

			brut	*net*	*Net space use*	**Needed emptyness**
Total m² area to cultivate :			20,000	18,000	90.0%	5%
Total week x m² :				936,000		

Duration culture and space:	**# weeks**	**# plants**	**# plants**	**# wkm²**	**Space use**					
Culture Phase			**per nm²**		**in nwkm²**	**in %**	**in m²**			
Phase 1 (Plug)	24	1000	350.0	2.9	68.6	8%	1,561			
Phase 2 (Growing 1)	18	1000	85.0	11.8	211.8	24%	4,820	9,883		
Phase 3 (Growing 2)	10	1000	65.0	15.4	153.8	18%	3,502			
Phase 4 (Cooling)	7	1000	45.0	22.2	155.6	18%	3,541			
Phase 5 (Flowering)	11	1000	45.0	22.2	244.4	28%	5,564	10,117		
Phase 6 (Finishing)	2	1000	45.0	22.2	44.4	5%	1,012			
Total	72				879		20,000			

			Loss	**5%**	**excl. loss**	**incl. loss**
# plants to deliver per year:					961,432	1,012,034
per week					18,489	19,462
per period (13x)					73,956	77,849
per month (12x)					80,119	84,336
# plants per m² per year					**48.1**	50.6